C1_AIB0326 · ISB ALP TEAM 10 · REVIEW 2

iADAS

Intelligent Anomaly Detection & Alerting System

Forecast-Based Retail Sales Anomaly Detection & Business Driver Investigation

SINCE REVIEW 1

Project Overview

Objective

Develop an AI-powered retail sales monitoring system that forecasts weekly sales, detects anomalies by comparing forecasted and actual sales, identifies key business drivers, and generates business-friendly alerts.

Changes Since Review 1

- Confirmed the **Online Retail II** dataset (Kaggle / UCI) as the primary data source
- Shifted from **statistical to forecast-based** anomaly detection for richer context
- Added **customer behavior anomaly detection** as an analytical layer

PROGRESS SNAPSHOT

Milestones & Deliverables

MILESTONE	DELIVERABLE	STATUS
Project Planning	Project Charter & Architecture	✅ Completed
Data Engineering	Cleaned Dataset & Weekly Aggregation	✅ Completed
AI Model	Prophet Forecasting Model	✅ Completed
Analytics	Forecast-Based Anomaly Detection	✅ Completed
Business Intelligence	Business Driver Investigation & Reports	✅ Completed
Documentation	GitHub Repository & README	✅ Completed
Final Enhancement	Dashboard & LLM Integration	🚧 In Progress

PHASE · PROTOTYPE VALIDATION & ENHANCEMENT

Current Status & Progress

Progress Since Review 1

- Functional end-to-end AI workflow implemented
- Core forecasting and anomaly detection modules completed
- Automated reporting and business investigation operational
- Dashboard and LLM integration planned for final phase
- On track for Review 3 final demonstration

~85%
Overall project completion

100%
Core AI pipeline complete

2
Workstreams in final phase

PROTOTYPE VALIDATED

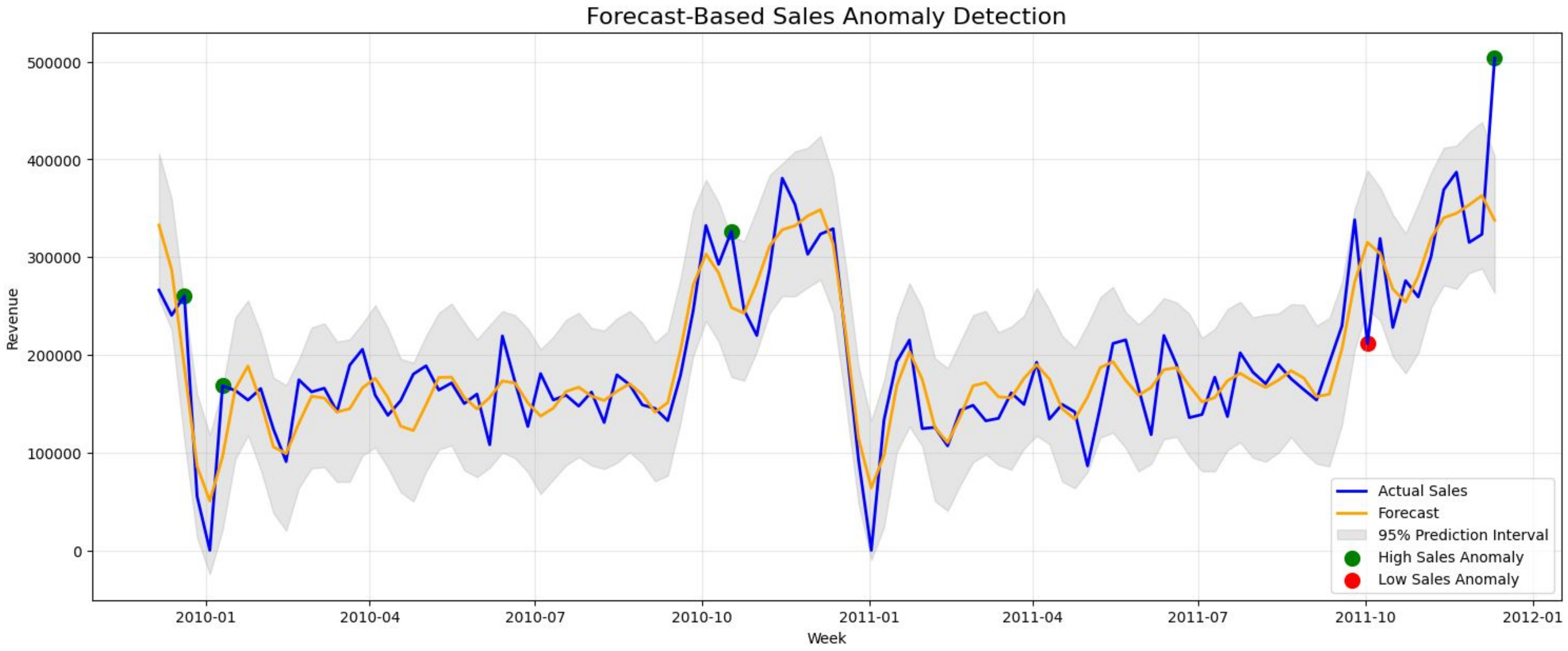
Working Prototype & Key Results

1.06M
Transactions

106
Weeks of Data

Meta Prophet
Forecast Model

5
Anomalies Detected



Successfully implemented and validated an end-to-end AI workflow for forecast-based retail sales anomaly detection using the **Online Retail II** dataset.

CHALLENGES HANDLED & WATCH ITEMS

Challenges & Risks

CHALLENGE	MITIGATION
Limited historical data (106 weekly observations) reduced forecasting complexity.	Selected **Meta Prophet**, which performs well with smaller time-series datasets.
Static thresholds generated excessive false alerts during initial experimentation.	Implemented **forecast-based anomaly detection** using Prophet's prediction intervals.
Dataset lacked business context (promotions, holidays, inventory events).	Developed **Business Driver Investigation** to identify key products, customers, and growth drivers.
Technical outputs were difficult for business users to interpret.	Generated **executive-ready reports** with business-friendly insights and recommendations.

Project Risks

Limited historical data may affect forecast performance during unusual business events.

LLM integration requires additional prompt engineering and testing before deployment.

PLAN VS. ACTUAL

Timeline

PHASE	REVIEW 1 PLAN	CURRENT STATUS	ADJUSTMENT
Project Charter & Problem Definition	✅ Planned	✅ Completed	No change
Data Collection & Preparation	Week 2	✅ Completed	Expanded data cleaning and feature engineering
Forecasting Model	Week 3	✅ Completed	Standardized on Meta Prophet after evaluation
Anomaly Detection	Week 4	✅ Completed	Enhanced with prediction interval-based detection
Business Investigation & Reporting	Week 5	✅ Completed	Added automated Business Driver Investigation and executive reports
Dashboard & LLM Integration	Final Phase	🚧 In Progress	Shifted to Review 3 to prioritize a stable core AI pipeline

THE ROAD TO REVIEW 3

Next Steps & Action Plan

NEXT STEP	EXPECTED COMPLETION
Develop Interactive Streamlit Dashboard	Before Review 3
Integrate LLM for AI-generated Business Insights	Before Review 3
Enhance Executive Reports & Recommendations	Before Review 3
End-to-End Testing & Performance Validation	Before Final Review
Prepare Final Presentation & Live Demo	Final Presentation

Action Plan · Focus for the next phase is to transform the current analytics prototype into an interactive AI decision-support system through dashboard development, LLM integration, and end-to-end demonstration.

WRAPPING UP

Conclusion & Summary

Key Achievements

- Built a functional end-to-end AI prototype for retail sales anomaly detection
- Implemented forecasting, anomaly detection, and automated business investigation
- Generated executive-ready reports to support business decision-making
- Established a scalable foundation for dashboard and LLM integration

Final Phase

1. **Interactive Dashboard**
   Real-time anomaly visualization and exploration
2. **Conversational AI Assistant**
   LLM-powered natural language business insights
3. **Live Demonstration**
   End-to-end Review 3 presentation

Thank You